

# A Review of Active Collaborative Perception and Planning for Heterogeneous UAV-UGV Systems

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**Abstract**—Unmanned aerial vehicles (UAVs) and unmanned ground vehicles (UGVs) offer complementary capabilities—wide-area aerial surveillance versus high-payload ground operation—making their coordination a compelling paradigm for complex real-world missions. Yet deploying such heterogeneous systems remains challenging owing to significant perceptual viewpoint discrepancies, tightly coupled decision-making, and constrained communication and computation budgets. Existing surveys address perception, planning, or control in isolation, leaving a critical gap in the systematic analysis of the full perception–planning–decision–control chain. This paper presents a structured review of UAV-UGV collaborative systems, covering cooperative architecture design, multi-source sensor fusion, active perception strategies, multi-agent planning and decision-making, and motion control. We further analyze key deployment bottlenecks under dynamic real-world conditions and discuss how large language models (LLMs) and embodied intelligence are poised to reshape the architectural paradigm of future air-ground systems.

**Index Terms**—air-ground heterogeneous collaboration; UAV-UGV; active perception; multi-agent reinforcement learning; embodied intelligence

## I. INTRODUCTION

The complementary nature of UAVs and UGVs—aerial breadth versus ground precision—has established air-ground heterogeneous systems as a key enabler for missions ranging from disaster response to precision agriculture. UAVs provide rapid wide-area situational awareness but are limited in payload and endurance; UGVs deliver robust ground manipulation and transportation yet suffer from terrain-constrained visibility. Their integration enables a “global perception, local execution” closed loop that neither platform can achieve alone [1].

Despite this promise, realizing efficient collaboration faces three fundamental challenges. First, **perceptual heterogeneity**: UAV top-down imagery and UGV frontal LiDAR/depth cameras differ substantially in scale, resolution, and geometry, complicating data fusion and shared environment representation [2]. Second, **active perception requirements**: passive, fixed-path sensing is insufficient; true intelligent coordination demands that each agent autonomously plan optimal observation viewpoints to maximize information gain, imposing tight perception–planning coupling [3]. Third, communication bandwidth constraints, real-time decision-making requirements under dynamic uncertainty, and the sim-to-real transfer gap collectively hinder reliable field deployment [4].

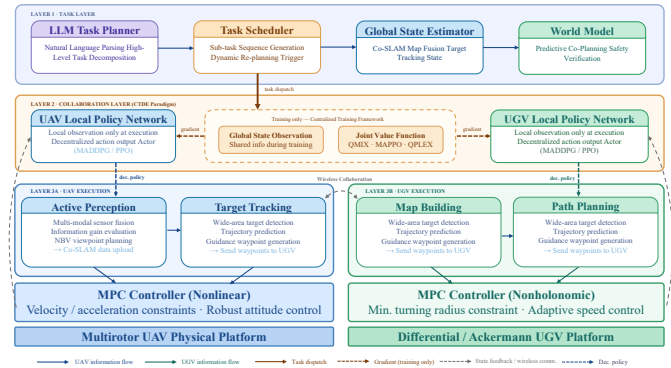


Fig. 1. Overall architecture for active collaborative perception and planning in heterogeneous UAV-UGV systems.

Unlike prior surveys that address individual modules in isolation, this paper surveys the complete cooperation chain spanning architecture, perception, planning, control, and foundation-model directions, as illustrated in Fig. 1.

## II. COOPERATIVE ARCHITECTURE AND SYSTEM MODELING

### A. Platform Dynamics and Kinematic Constraints

The core asymmetry between UAV and UGV stems from their contrasting dynamics. A typical multi-rotor UAV enjoys 6-DOF agile motion governed by Newton–Euler equations [5], enabling hovering and rapid maneuvering. Most UGVs, by contrast, are subject to **nonholonomic constraints**: differential-drive models prohibit lateral motion, while Ackermann-steering models impose a minimum turning radius [6]. This kinematic mismatch creates a hard physical barrier for joint motion planning—3D UAV trajectories that violate UGV nonholonomic constraints cannot be directly executed—requiring cooperative planners to explicitly incorporate heterogeneous motion constraints into their formulations.

A notable special case is the **tethered system** [7], where a physical cable connects UAV to UGV, enabling continuous power supply and significantly extending UAV endurance. The cable simultaneously introduces additional tension constraints and collision-avoidance complexity, converting the relatively

independent dual-agent problem into a strongly coupled co-control problem—a canonical tradeoff between endurance and kinematic freedom.

### B. Cooperative Control Architectures

Cooperative control architecture determines information flow, decision mode, and overall system performance. Three paradigms dominate current research. **Centralized architectures** aggregate all sensing data at a single compute node for global planning, yielding theoretically optimal solutions but suffering from poor scalability, heavy communication overhead, and a single point of failure [8]. **Decentralized architectures** grant each agent autonomous decision-making from local observations, offering strong robustness and scalability, but local information may lead to suboptimal outcomes [9]. **Hierarchical architectures** balance the two by decoupling high-level task allocation from low-level trajectory tracking [6].

The rise of MARL has elevated **Centralized Training with Decentralized Execution (CTDE)** to the dominant cooperative framework [9], [10]: global state is exploited during training while each agent executes using only local observations at runtime. Value-decomposition algorithms exemplified by QMIX [10] make this paradigm particularly effective for tight cooperative tasks such as coordinated exploration and target encirclement. The effectiveness of CTDE under partial observability has been further validated in asynchronous multi-agent settings [11], as illustrated in Fig. 2.

## III. ACTIVE COLLABORATIVE PERCEPTION

### A. Multi-Source Heterogeneous Sensor Fusion

The perceptual advantage of air-ground systems arises from the complementarity of UAV top-down and UGV close-range sensing, yet this same complementarity creates the primary data fusion barrier [2]. Conventional geometry-based feature matching methods (SIFT, ORB) routinely fail on top-down/frontal image pairs due to sparse feature correspondence. Research has consequently shifted toward **deep learning-based adaptive feature fusion**, where attention mechanisms dynamically weight contributions from different viewpoints, establishing cross-modal semantic associations without requiring precise pixel-level alignment.

At a higher level, **Collaborative SLAM (Co-SLAM)** enables UAV and UGV to independently build local submaps subsequently merged via server-side global consistency optimization [12], providing heterogeneous agents with a shared, accurate environmental representation that underpins the viewpoint planning in Section III-B.

### B. Active Perception Strategies: From Information-Driven to Task-Driven Viewpoint Planning

The field has evolved from model-based optimization toward end-to-end learning. Classical information-theoretic methods—frontier exploration and Next-Best-View (NBV) planning [13]—maximize an information-gain function over candidate viewpoints but depend on computationally expensive

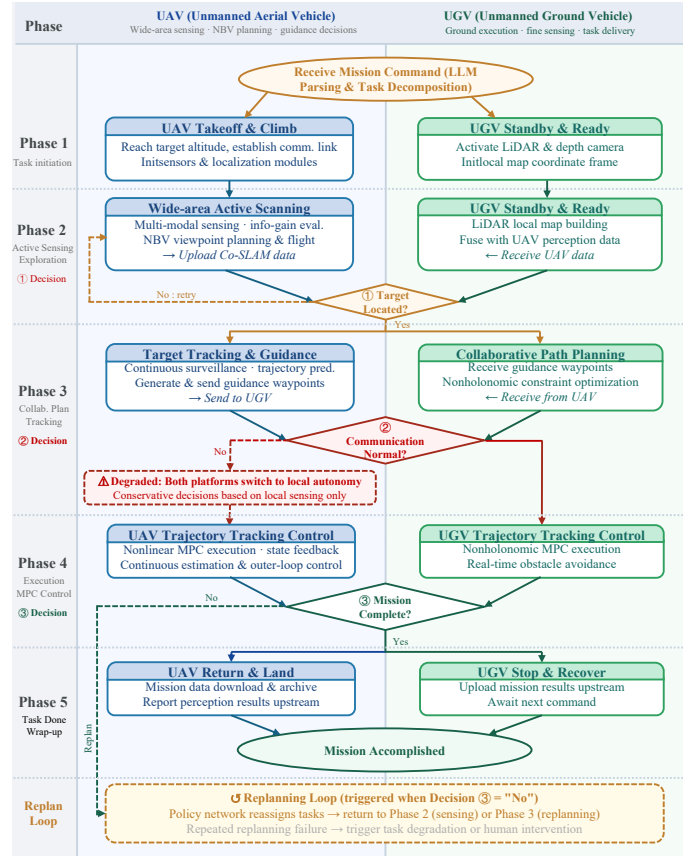


Fig. 2. UAV-UGV active collaborative perception and planning task execution flow.

ray-casting that limits real-time performance. Active Neural SLAM [14] replaced hand-crafted heuristics with end-to-end deep reinforcement learning, achieving continuous exploration in highly dynamic environments.

Vutetakis et al. further proposed an active perception network that predicts information gain directly from local viewpoint features, entirely bypassing ray-casting and enabling non-myopic online exploration [15]. For air-ground cooperative exploration, Dechicha et al. combined UAV-based wide-area frontier scouting with UGV-based local NBV refinement [7], embodying an effective heterogeneous division of labor: the UAV reduces global uncertainty while the UGV extracts fine-grained local detail. Despite their promise, learning-based policies remain sensitive to observation noise, motivating hybrid model-data approaches for deployment.

### C. Cooperative Target Tracking

Cooperative target tracking exemplifies the closed-loop synergy between a “sensing eye” (UAV) and an “executing hand” (UGV). In the classical UAV-guided paradigm, the UAV tracks targets from altitude and feeds interception waypoints to the UGV in real time. Multi-agent reinforcement learning has driven major advances: Munasinghe et al. demonstrated that proficiency-constrained MARL yields significantly higher cooperative tracking success rates in dynamic vehicle pursuit [8];

Zhao et al. trained UAV formations via MADDPG [16] to achieve persistent multi-target tracking in complex environments [17], with a graph neural network extension further scaling cooperative perception to large heterogeneous swarms. The target state estimates produced by this perception stack directly feed the collaborative planning module discussed next.

#### IV. COLLABORATIVE PLANNING AND DECISION-MAKING

Collaborative planning translates perception outputs into coordinated action sequences under kinematic, dynamic, and task constraints [18]. Table I compares representative approaches spanning rule-based optimization and end-to-end learning.

##### A. From Explicit Optimization to Learned Decision-Making

Early methods relied on rule-based optimization. Market auction mechanisms allocate tasks through distributed bidding but provide no optimality guarantees [19]. MILP yields globally optimal solutions but scales exponentially, precluding real-time replanning [6].

Deep MARL offers a fundamentally different paradigm by learning state-to-action mappings through trial-and-error interaction. **QMIX** employs a monotonic mixing network to factorize the global action-value function into per-agent local utilities, enabling CTDE and excelling on purely cooperative tasks [10]. **MADDPG** adopts a centralized critic–decentralized actor framework, effectively handling mixed cooperative-competitive settings [16]. **MAPPO** [20] achieves stable, sample-efficient cooperative training, while **QPLEX** [21] introduces a duplex dueling network to overcome QMIX’s monotonicity constraint.

Despite these advances, MARL deployment in air-ground systems faces persistent challenges: sparse rewards impede learning; sim-to-real transfer degrades under perceptual noise and model mismatch [4]; and credit assignment among heterogeneous agents in non-stationary environments remains an open problem [24].

#### V. COOPERATIVE MOTION TRACKING CONTROL

**Model Predictive Control (MPC)** has emerged as the preferred solution for air-ground trajectory tracking owing to its native handling of multi-input multi-output systems and hard constraints. By solving a finite-horizon optimal control problem at each timestep in a receding-horizon fashion, MPC explicitly enforces velocity, acceleration, and inter-platform collision-avoidance constraints. Nan et al. demonstrated nonlinear MPC’s effectiveness for quadrotor fault-tolerant control [22]; Romero et al. extended model predictive contouring control to time-optimal flight [23], confirming MPC’s engineering viability for high-precision cooperative maneuvering. The heavy online computation of nonlinear MPC—particularly acute in multi-platform synchronous control—motivates ongoing work on lightweight formulations and neural network approximations.

While classical MPC addresses execution-level robustness effectively, task-level generalization across open-world scenarios remains beyond its scope, motivating the foundation-model directions discussed next.

#### VI. FUTURE DIRECTIONS: FOUNDATION MODELS AND EMBODIED INTELLIGENCE

Current air-ground systems have matured considerably across the perception-to-control stack, yet fundamental limitations remain in task generalization and long-horizon reasoning under open-world conditions. The integration of LLMs and world models is reshaping the architectural paradigm toward general-purpose embodied intelligence.

##### A. LLMs as High-Level Task Planners

LLMs enable air-ground systems to accept natural language directives and decompose them into executable subtask sequences. Given “survey post-fire damage on the east side,” an LLM decomposes this into UAV infrared scanning followed by UGV close-range inspection [25]. The ELLMER framework shows that LLMs with retrieval-augmented generation can dynamically revise plans across long, unpredictable task horizons [26]. RT-2 vision-language-action models further transfer web-scale knowledge to robotic manipulation, opening a pathway for open-world semantic reasoning in air-ground systems [27].

##### B. World Models and Embodied Intelligence

A world model serves as an agent’s internal simulator of environment dynamics, enabling lookahead planning and safety verification by imagining the consequences of cooperative strategies before real-world execution—dramatically reducing trial-and-error costs [28]. DreamerV3 demonstrates the feasibility of learning generalizable world representations across diverse physical domains [28], with considerable value for air-ground applications. Looking ahead, systems that fuse LLM task planning, world-model causal prediction, and foundation-model perception will evolve from fixed-script automation tools into embodied collaborative agents capable of understanding intent and autonomously devising cooperative strategies [29].

#### VII. CONCLUSION

This paper has reviewed active collaborative perception and planning for heterogeneous UAV-UGV systems along the full perception–planning–decision–control chain. Three takeaways emerge from this analysis. At the architectural level, CTDE has become the dominant cooperative framework, with hierarchical MARL representing the most engineering-viable path to distributed intelligence. In the perception domain, task-driven active information acquisition has superseded passive fusion, though hybrid model-learning approaches remain necessary for reliable deployment. On the planning side, learned decision-making increasingly outperforms classical optimization, but the sim-to-real gap demands co-design of data-driven policies with model-based constraints. Bridging this gap, alongside advances in cross-modal alignment and emergent coordination in large-scale swarms, constitutes the central challenge for the field. Looking further ahead, LLMs and world models offer a transformative vision: air-ground systems that transcend task-specific automation to become

TABLE I  
COMPARISON OF MAIN COLLABORATIVE PLANNING AND DECISION-MAKING ALGORITHMS

| Algorithm      | Type                        | Applicable Scenario           | Core Advantage                          | Main Limitation                          | Air-Ground Application            |
|----------------|-----------------------------|-------------------------------|-----------------------------------------|------------------------------------------|-----------------------------------|
| MILP / ILP     | Combinatorial Opt.          | Static task allocation        | Global optimality; explicit constraints | Exponential complexity; poor scalability | Multi-target assignment [18]      |
| Market Auction | Game Theory                 | Dynamic task allocation       | Decentralized; scalable                 | No optimality guarantee                  | Multi-agent negotiation [19]      |
| MADDPG         | Policy-Gradient MARL (CTDE) | Mixed cooperative-competitive | Rich policy expressiveness              | Training instability                     | Target tracking [8], [17]         |
| QMIX           | Value-Dec. MARL (CTDE)      | Purely cooperative            | Efficient; joint optimality             | Monotonicity constraint                  | Collaborative exploration [10]    |
| MAPPO          | PPO-MARL (CTDE)             | Cooperative tasks             | Stable; sample-efficient                | Credit assignment difficulty             | Multi-UAV formation [20]          |
| QPLEX          | Duplex Value-Dec. (CTDE)    | Cooperative tasks             | Overcomes QMIX monotonicity             | Complex architecture                     | Complex coordination [21]         |
| MPC            | Model Predictive Control    | Trajectory + constraints      | Explicit constraint handling            | Heavy online computation                 | Collaborative tracking [22], [23] |

embodied collaborative agents capable of natural-language instruction following, physical reasoning, and long-horizon planning.

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